CASTING A BIG NET: Hunting At Scale

CROWDSTRIKE

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- Principal Consultant, CrowdStrike
- Previously Fortune 10, Government, Energy, Manufacturing, and Telecommunications industries
- Enjoys *sleep*, family time, craft beer, bourbon, college football, and fighting nation state adversaries

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- Senior Consultant, CrowdStrike
- Previously Ernst & Young
- Crossfit Flounder
- #PCMasterRace

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Agenda

- Hunting Essentials
- Our Approach
- Collections with CrowdResponse
- Rapid Analysis with Splunk
- Demo
- Key Takeaways / Q&A

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HUNTING: WHAT IT IS NOT

- Not a new concept
- Not alert driven
- Not a tool or product
- Not standardized
- Not a silver bullet

What Sales thinks hunting is

Image Credit: hxxps://www.pinterest.com/monsterstein1/army-guns/

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HUNTING: WHAT IT IS

- The Ultimate DFIR Question
- Proactive
- Component of Active Defense
- Assumption of Breach
- Exploring your Ignorance*
  - Select data sources
  - Baseline behaviors
  - Identify anomalies
  - Learn> Apply> Detect> Repeat

What CISOs think hunting is

Image Credit: hxxps://www.northropgrumman.com/

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HUNTING: HOW ITS DONE

- Approaches
  - Objective Driven
  - Intelligence Driven
  - Anomaly Driven
- Frequency
  - Ad-hoc> Project-based> Continuous
- Data analysis techniques
  - Rare – Least Frequency of Occurrence
  - Variance - Standard Deviation
  - New – 1st seen on Network/Host
  - Sampling – Data not normally collected
  - Temporal – Timing oddities

What hunting really is

Image Credit: hxxps://home.bt.com/

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HUNTING: WHO & WHY

Strengths (Who)
Incident Response
Network Visibility
Host Visibility
Centralized Logging

Weaknesses (Why)
Prevention
Early Detection
Limited Experience
Budget

Opportunities (Why)
Loss Avoidance
Risk Assurance
Improve Detection & Response
Job Enrichment

Threats (Why)
Organized eCrime
Malicious Insiders
Hactivists
Nation State Adversaries

Image Credit: hxxps://www.piedpiper.com/

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HUNTING: OUTPUTS

- Report & Debrief
- Increased Security Assurance
- Increased Business Intelligence
- Actionable Monitoring & Detections
- Loss Avoidance
- Staff Training & Enrichment

Image Credit: hxxps://imgflip.com/

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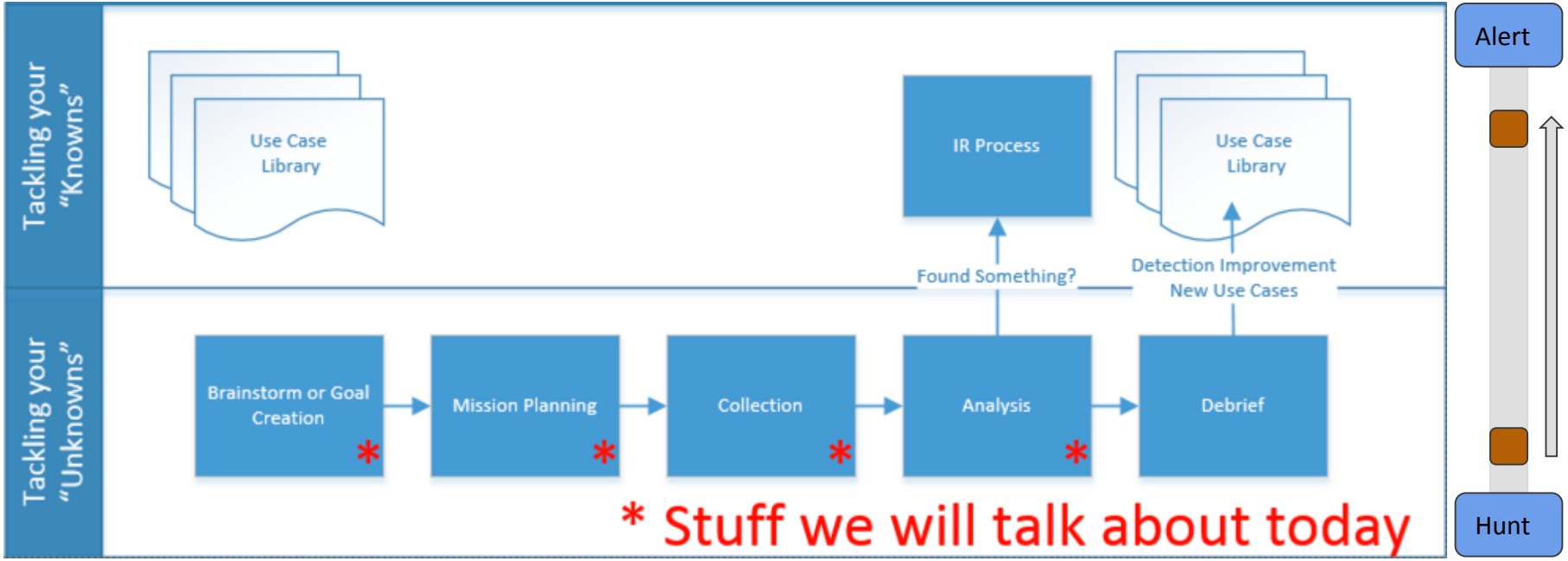
HUNTING: THREAT DETECTION FRAMEWORK



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HUNTING: EXAMPLE HUNT

Goal Creation
- What data are you currently collecting?
- What data can you collect?
- How much data will be manageable?

Mission Planning
- How are you going to collect data?
- What tools are needed to collect?
- How long will collection take?

Collection
- Rubber meets the road
- Putting my toolset from to use

Analysis
- Is my analysis environment operating?
- Is my data coming in a format I can analyze?

Debrief
- Findings?
- Lessons Learned?
- Can this be tuned and turned into an alert?

Goal Creation
- What programs have established persistence in my environment?

Mission Planning
- PSEXEC, CrowdResponse, and an open share

Collection
- CAB & XML collection
- Splunk ingestion of processed CSV

Analysis
- Splunk queries for least frequency
- Analyst knowledge of other indicators

Debrief
- 10 binaries to analyze and follow up on
- 200 machines with commodity infections

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MISSION PLANNING: COLLECTING WITH CROWDRESPONSE

Author
- Robin Keir

Components
- CrowdResponse.exe - Windows Console Application
- CRConvert.exe - Output Parser

Output
- XML -> CSV, TSV, HTML, Plain Text

CrowdResponse Deployment
- Targeted Command Line
- External Config

Deployment Options
- Psexec
  - see SANS Blog on protecting your privileged credentials (IPC$ auth)
- Powershell
  - https://github.com/n3l5/irCRpull
- GPO
- SCCM

Collection/Analysis Platform
- Splunk preferred

WARNING: CrowdStrike is not responsible for your use (or misuse) of the tools listed

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MISSION PLANNING: COLLECTING THE CROWDSTRIKE WAY

Our Lessons Learned:

- Frictionless deployment & collection
- Persistence vs Self Cleaning
- Establish a centralized intake server to receive files
- Intake server decompresses and crconverts data into Splunk accessible directory
- Large collections can create havoc, be prepared to adjust if needed
- Multiple indexes can separate hunts
- Scripts to handle duplicates
- IOPS reign supreme
- Temporary open share for ad-hoc collections

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CROWDRESPONSE: FILES

Artifact Refresh:

- Filesystem Metadata about files
- $MFT or FAT
- MACE/MACB timestamps

```
@DirList [-chmqrstx] [-f <mask>] [-ei <filter>] [-p <depth>] [-z <size>] <dir>

 -c              - Verify and display digital signature information
 -e <filter>     - Path exclusion regex filter pattern
 -f <mask>       - File name mask. Default is "*.*"
 -h              - Do SHA256 hashes
 -i <filter>     - Path inclusion regex filter pattern
 -m              - Do MD5 hashes
 -p <depth>      - Recursion depth (0 = top level only, 1 = 1 level down etc)
 -q              - Perform "quick" hash of only the 1st 512 bytes of the file
 -r              - Display application resource information
 -s              - Recursive listing
 -t              - Display file create, modified and accessed times
 -x              - Only process Windows executable (PE) files
 -z <size>       - Don't hash files greater than this size in megabytes
 <dir>           - Directory to start listing from
```

Example:

Directory listing and all sub-directories looking for executables (exe)

```
@DirList "%windir%" -s -f *.exe
```

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CROWDRESPONSE: REGISTRY

Artifact Refresh:

- Data files called “HIVES”
- Key and Value Pairs
- Multiple Hives for User/System

Windows Reg HIVES are a treasure trove of data

- UserAssist
- MRU
- **ShimCache**
- **Services**
- **And much more!**

```
@RegDump [-dstT] [-r <remote>] <reg key>

-d            - Nested output format
-r <remote>   - Connect to remote registry at [remote]
-s            - Recursive dump
-T            - Output additional wide character text version of binary data
-t            - Output additional ASCII text version of binary data
<reg key>     - Registry key to start dump from
```

```
@RegFile [-chmqrst] [-z <size>] <reg key>

-c            - Verify and display digital signature information
-h            - Do SHA256 hashes
-m            - Do MD5 hashes
-q            - Perform "quick" hash of only the 1st 512 bytes of the file
-r            - Display application resource information
-s            - Recursive dump
-t            - Display file create, modified and accessed times
-z <size>     - Don't hash files greater than this size in megabytes
<reg key>     - Registry key to start dump from
```

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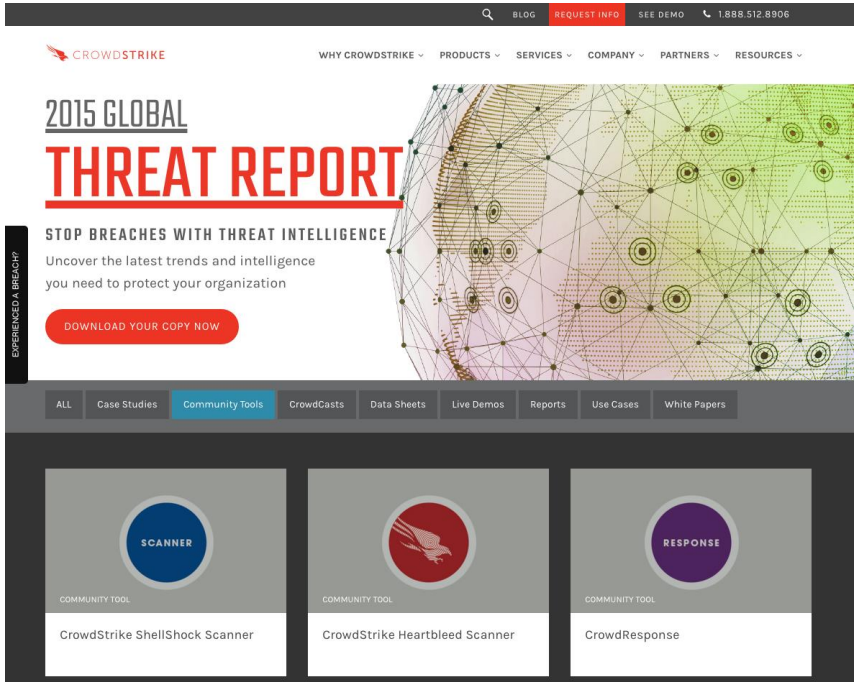
CROWDRESPONSE: INTRODUCING TASKS

Introducing: **CrowdResponse Tasks**

- Available after #ThreatHuntingSummit on www.crowdstrike.com/resources

New module to collect Windows Scheduled Tasks information

- Processes Task Scheduler 1.0 and 1.2 (vista+)
- Task name
- Application
- Parameters
- Working Directory
- And more!



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CROWDRESPONSE: TASKS

Artifact Refresh:
Another persistence method and possibly privilege escalation leveraging the Windows Task Scheduler

Can be found in:

- C:\Windows\Tasks\ (XP - Windows Job Format)
- C:\Windows\System32\Tasks\ (Win7+ - XML)
- C:\Windows\SysWow64\Tasks\ (Win7+ - XML)

```
C:\> schtasks
C:\> at
```

```
@Tasks
```

…yes that's it

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CROWDRESPONSE: UPDATED MODULES

The new CrowdResponse also features updates to your favorite modules!

- @Mal now includes option to detect DLL Hijacking
  - Checks for DLL name conflicts found in $PATH
  - Checks for DLL name conflicts NOT found in $PATH
  - Checks svchost.exe processes for same DLL name loaded in different path
- Additional bug fixes and improved processing with CRConvert

WARNING: errors can occur if using old CRConvert to parse new collections

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ANALYSIS: THE NEXT STEP

Collection is mostly solved...but some problems remain

- Unsupported Operating System (*NIX, OS X, iOS, Android, ChromeOS, etc)
- Offline hosts
- Network bandwidth (Memory, Disk Images, etc)
- Legal (EU, Works Councils, etc)

Establishing a reliable analysis platform is a much bigger challenge initially

- Easy to scale
- Fast, repeatable searches
- Automatic data ingestion
- Automatic field parsing
- Easy data manipulation and statistical functions
- Timelining

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ANALYSIS:
RAPID ANALYSIS W/ SPLUNK

Splunk vs ELK vs Other

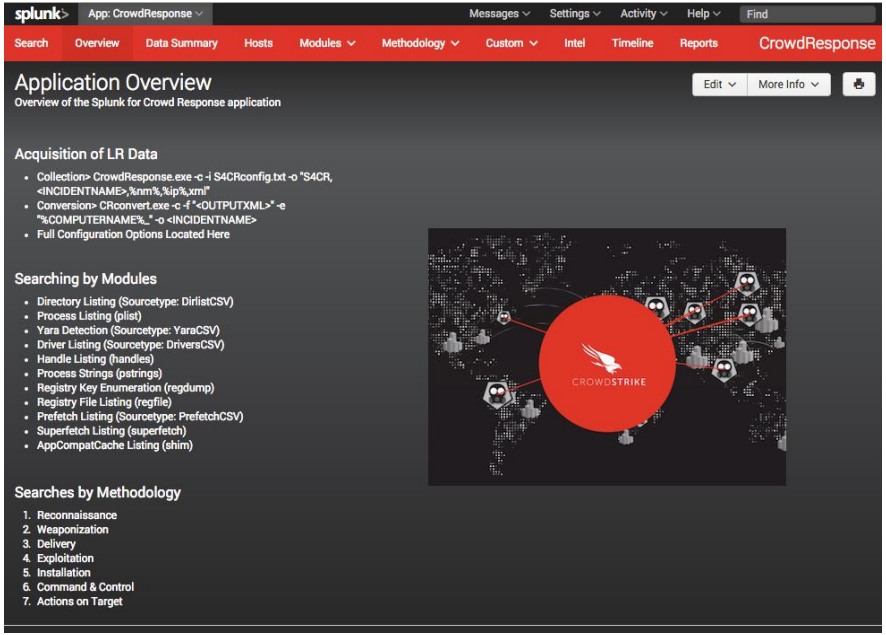
- Not traditional SIEM-like data ingestion
- Live Response Centric

Splunk applications

- Versioning - 6.x
- Advanced XML or HTML, CSS, Javascript

Application goals

- Faster searching
- Repeatable workflow
- Easy Pivoting



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ANALYSIS: RAPID ANALYSIS W/ SPLUNK

Module-based hunting

splunk> App: CrowdResponse

Messages Settings Activity Help Find

Search Overview Data Summary Hosts Modules Methodology Custom Intel Timeline Reports CrowdResponse

DirectoryList

DirectoryList
DriverList
HandleList
MFTDeleted
Prefetch
ProcessList
ProcessStrings
RegistryDump
RegistryFiles
ShimCache

Edit More Info

All time

HostName

All Hosts

Submit

3m ago

Known Bad Filenames

HostName	Files	MD	
ProdAPP2	cmd.exe	07	s (x86)\Business Objects\common\4.0\setup\InfoZip\zip.exe
	win32k.sys	0b	s\CommVault\Simpana\Base\zip.exe
	zip.exe	19	s\VMware\VMware Tools\zip.exe
		1d7edead6891810bcf8566b2319a8b11	C:\Users\755921-MRebhan\AppData\Local\Temp\zip.exe
		275d3946b0ec22ba13fe299e97abf606	C:\Windows\SysWOW64\cmd.exe
		2e54977ced249debfbbfb205dd27c89c	C:\Windows\System32\cmd.exe
		32b9233c4a7a4a546436a26e28ff8aa6	C:\Windows\System32\win32k.sys
		338e48ab7810e1b223dfecd82c44f5a3	C:\Windows\winsxs\amd64_microsoft-windows-
		42a88ecf903bfe11411d188dce830e84	commandprompt_31bf3856ad364e35_6.1.7601.17514_none_e932cc2c30fc13b0\cmd.exe
		511166d3f5d7eba36de48c4f5e195886	C:\Windows\winsxs\amd64_microsoft-windows-
		523b9b64f2b6c630a2e0a87116c05f12	win32k.31bf3856ad364e35.6.1.7601.17514.none.17197f29b1265401\win32k.sys

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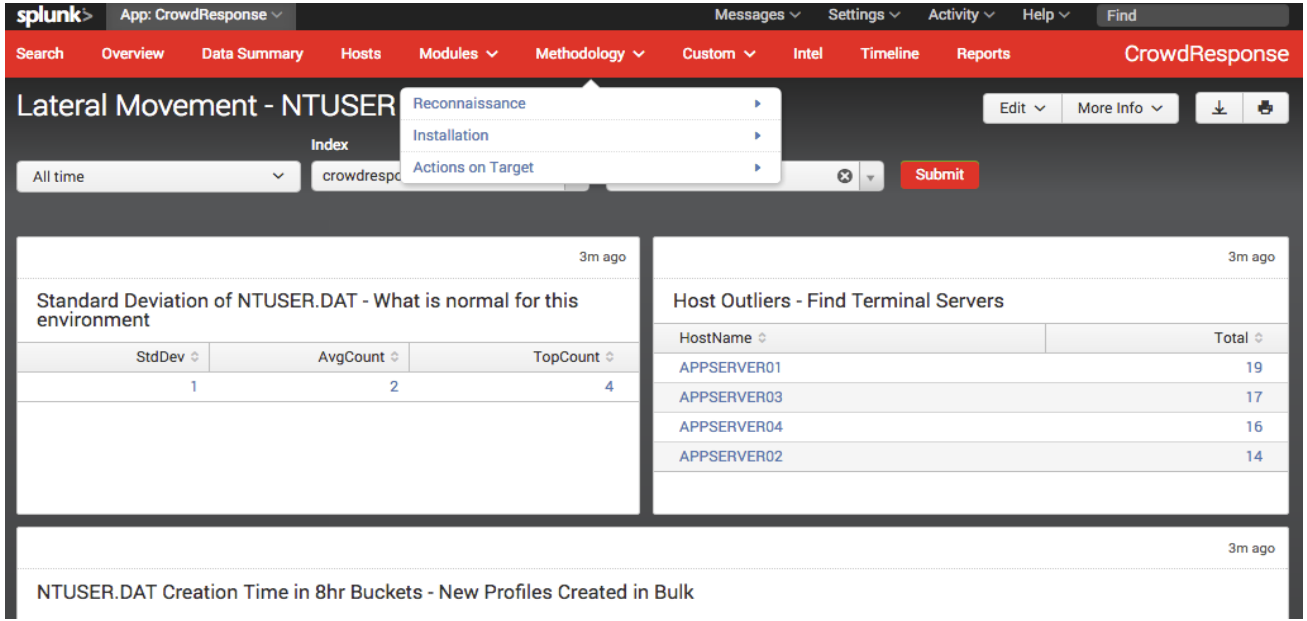
ANALYSIS:
RAPID ANALYSIS W/ SPLUNK

Methodology-based hunting



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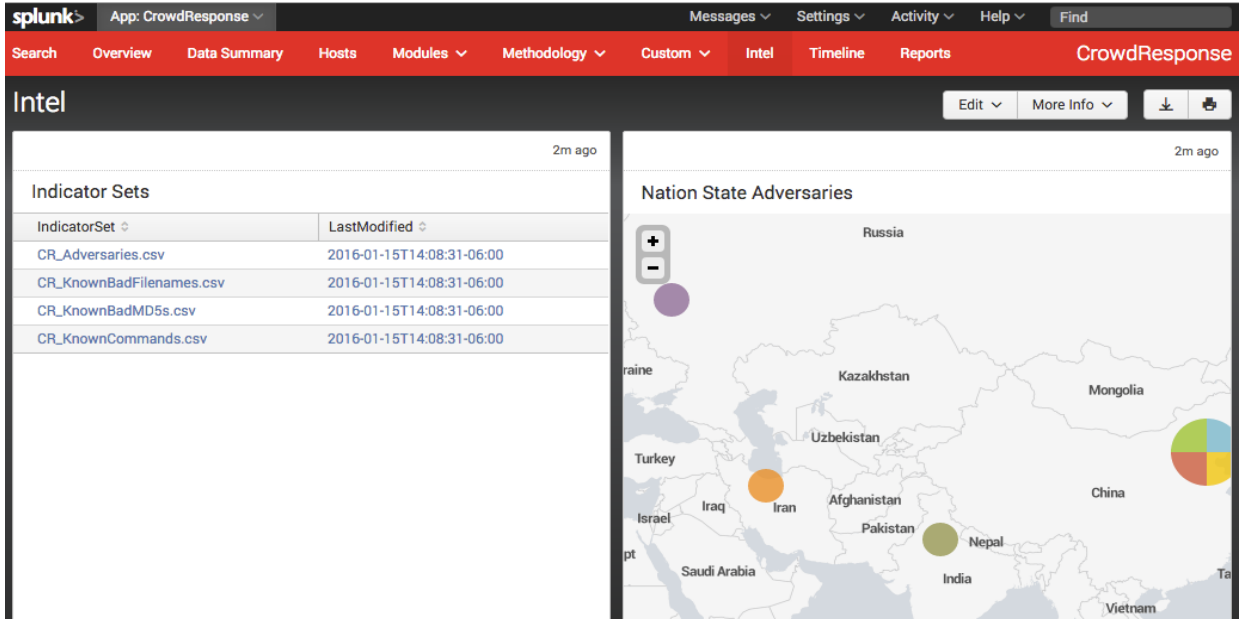
ANALYSIS:
RAPID ANALYSIS W/ SPLUNK

Indicator Management Lite



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DEMO:
HUNTING W/ CROWDRESPONSE

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DEMO:
HUNTING RECAP

Let's recap what we observed:

- Indicator hits for DEEPPANDA SAKULA malware and credential dumper
- Persistence for a netcat listener
- "At" Scheduled Tasks for a batch file
- Sticky keys debugger set to cmd.exe for the On-Screen Keyboard (osk.exe)
- ShimCache evidence of possible reconnaissance and exfiltration preparation

Not client data but all TTPs seen in CrowdStrike IRs

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HUNTING: **KEY TAKEAWAYS**

PLAN
Business & Threat Aligned Use Cases

Gain approval, organize resources, and EXECUTE

DELIVER tangible outputs to stakeholders

PROFIT from increased Security Assurance and Loss Avoidance

QUESTIONS?

Possibly Answers